Tu Yuang

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Robotics · Dexterous Manipulation · VLA Models · World Models · Reinforcement Learning

Education

Nanyang Technological University — Singapore
Master of Science in Mechanical Engineering — *Aug. 2025 – Present*

Shanghai Jiao Tong University — Shanghai, China
Bachelor of Engineering in Energy and Power Engineering — *Sep. 2021 – Jun. 2025*

Experience

Research Intern — Jun. 2026 – Present
*Agency for Science, Technology and Research (A*STAR) — ARTC* — *Singapore*

- Research attachment in robot learning at A*STAR ARTC, focusing on robotic manipulation and reinforcement learning

Algorithm Engineer (Internship) — Jun. 2024 – Sep. 2024
Shanghai Ranjing Technology Co., Ltd. — *Shanghai, China*

- Trained deep learning models and evaluated human skeleton keypoint recognition performance
- Developed rule-based algorithms for multi-exercise repetition counting and violation detection
- Designed image-processing methods for standing long jump distance estimation

Project Experiences

Fine-grained Manipulation of Humanoids Using VLA Model | *Master's Thesis* — Sep. 2025 – Present

- Collected multimodal data using MobileAloha and developed a fine-tuning pipeline for VLA models for fine-grained humanoid manipulation
- Reproduced and benchmarked pi0, ACT, and Diffusion Policy in the LeRobot framework, comparing their generalization, stability, transferability, and control responsiveness on real-world tasks

AI for Industry Challenge by Intrinsic (Part of Google) | *International Challenge* — Feb. 2026 – Present

- Built an end-to-end HIL-SERL pipeline for 6-DoF robotic cable insertion based on the AIC benchmark, spanning training, inference, and evaluation
- Developed the complete learning pipeline, including demonstration collection, reward classifier training, BC pretraining, RLPD policy optimization, human-in-the-loop refinement, and simulation evaluation

RoboMaster Competition (World Champion) | *Project Team Leader* — Dec. 2021 – Aug. 2023

- Led the research, development, and maintenance of the infantry robot's mechanical structure (2022)
- Led the quadrotor aerial robot project (2023): drone mechanical design in SolidWorks with lightweight, aerodynamic, and vibration optimizations based on simulation and testing
- Conducted visual positioning experiments with the Intel RealSense T265, implementing sensor-fusion algorithms for localization and debugging VIO/VINS-based visual positioning

Flight Control of Kilogram Quadrotor Cyclocopter | *Undergraduate Final Year Project* — Dec. 2024 – May 2025

- Developed a flight control algorithm for a quadrotor cyclocopter based on the PX4 platform, implementing PID control and a sensor-fusion EKF algorithm
- Built a Gazebo plugin modeling the physical effects of the rolling wing and completed the full simulation model; achieved hovering and attitude control in SITL and HITL simulation
- Designed the control-system hardware and electrical system of the aircraft and completed the hardware wiring

Modeling, Communication & Control of SCARA Robot | *Research Program (PRP)* — May 2022 – Sep. 2023

- Designed a machine-vision-based screw pre-tightening system for a SCARA robot, training a YOLOv8 model to detect screws and their types
- Developed a robot end-effector capable of automatic bit switching with a force-feedback control algorithm to determine tightening levels
- Performed forward/inverse kinematics analysis and completed robot modeling and simulation in MATLAB

Academic Awards

2025 — NTU School of MAE Graduate Scholarship (~USD 40,000, top 1%)

2025 — China College Students' Mechanical Engineering Competition, National Third Prize

2024 — SJTU Academic Progress Scholarship

2023 — RoboMaster National University Robot Competition, National First Prize

2022 — RoboMaster National University Robot Competition, National First Prize

2022 — SJTU C-class Scholarship